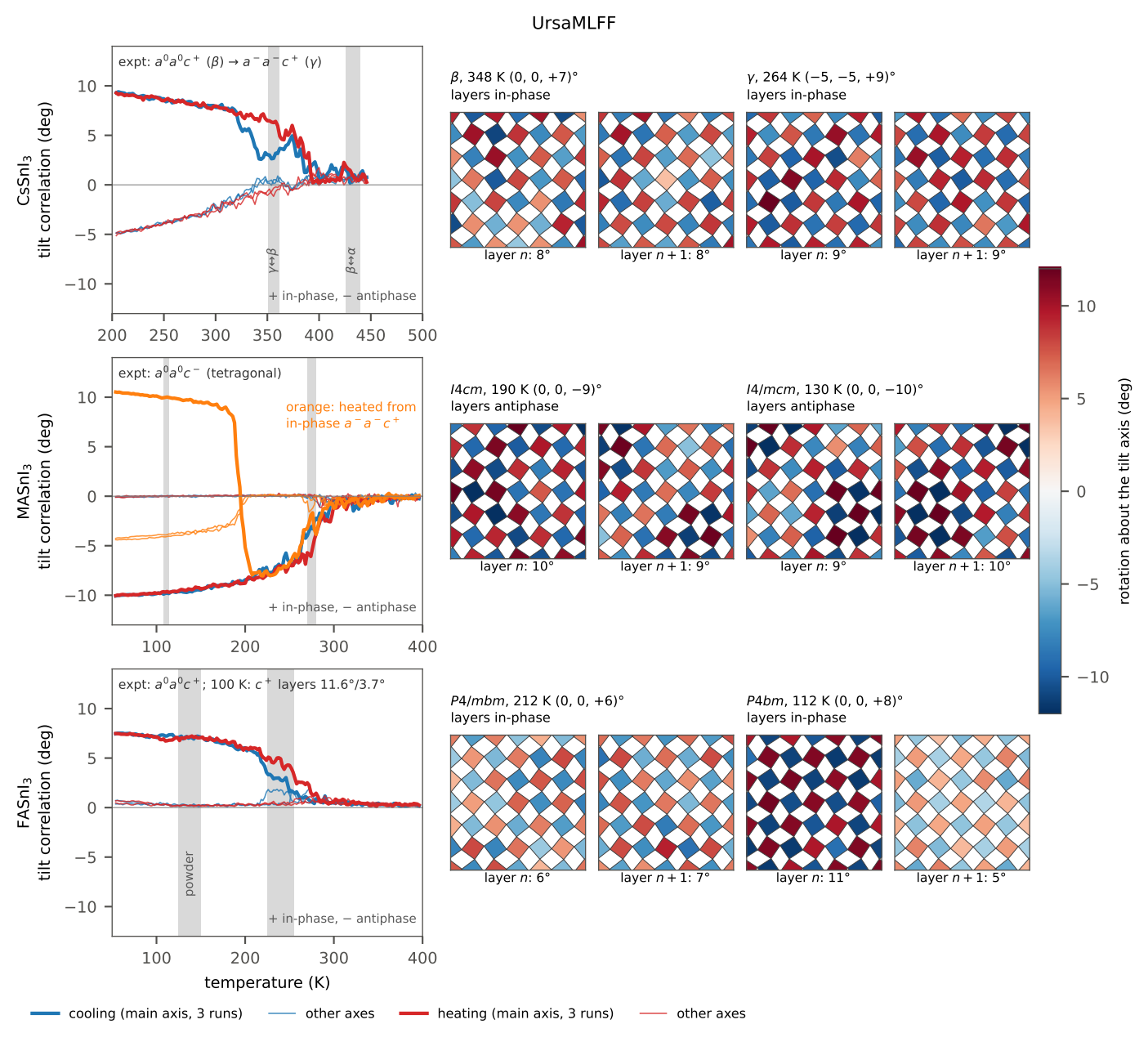

UrsaMLFF
CsSnI3 tilt correlation (deg)
expt: a0a0c+ (β) → a−a−c+ (γ)
γ↦β
β↦α
+ in-phase, − antiphase
β, 348 K (0, 0, +7)°
layers in-phase
γ, 264 K (−5, −5, +9)°
layers in-phase
layer n: 8°
layer n + 1: 8°
layer n: 9°
layer n + 1: 9°
MASnI3 tilt correlation (deg)
expt: a0a0c− (tetragonal)
orange: heated from in-phase a−a−c+
+ in-phase, − antiphase
I4cm, 190 K (0, 0, −9)°
layers antiphase
I4/mcm, 130 K (0, 0, −10)°
layers antiphase
layer n: 10°
layer n + 1: 9°
layer n: 9°
layer n + 1: 10°
FASnI3 tilt correlation (deg)
expt: a0a0c+; 100 K: c+ layers 11.6°/3.7°
powder
+ in-phase, − antiphase
P4/mbm, 212 K (0, 0, +6)°
layers in-phase
P4bm, 112 K (0, 0, +8)°
layers in-phase
layer n: 6°
layer n + 1: 7°
layer n: 11°
layer n + 1: 5°
rotation about the tilt axis (deg)
temperature (K)
cooling (main axis, 3 runs)
other axes
heating (main axis, 3 runs)
other axes